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Apparatus and method for manufacturing display device using ejection head ejecting adhesive material

Abstract

An apparatus and method for manufacturing a display device are provided. The apparatus for manufacturing the display device, includes: a stage, and an ejection head including an ejection outlet, which is open upward in a first direction opposite to a direction of gravity. The ejection head is configured to move in a second direction, which intersects the first direction, and disposed under the stage in the direction of the gravity. The ejection head includes a body part, which contains an adhesive material, and a stirring part, which is disposed in the body part and includes a roller, which rotates a predetermined number of times per unit time to control viscosity of the adhesive material.

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Background/Summary

(1) This application claims priority to Korean Patent Application No. 10-2021-0076752, filed on Jun. 14, 2021, and all the benefits accruing therefrom under 35 U.S.C. § 119, the content of which in its entirety is herein incorporated by reference.

BACKGROUND

1. Field

(2) The present disclosure relates to an apparatus and method for manufacturing a display device.

2. Description of the Related Art

(3) A display device is becoming more important with developments in multimedia technology. Accordingly, various display devices such as an organic light-emitting diode (“OLED”) display device, a liquid crystal display (“LCD”) device, and the like have been used. The application of such display devices has been diversified, ranging from various electronic devices such as televisions (“TVs”), smartphones, smartwatches, or tablet personal computers (“PCs”).

(4) Typically, the display device may include a cover window to protect a display panel. The cover window may be attached to the display panel via an optically clear adhesive (“OCA”). The OCA may be applied onto the display panel with the use of inkjet equipment or a roller.

SUMMARY

(5) Embodiments of the present disclosure provide an apparatus and method for manufacturing a display device, which can respond to various planar shapes of display panels and can apply an adhesive member to a uniform thickness.

(6) However, embodiments of the present disclosure are not restricted to those set forth herein. The above and other embodiments of the present disclosure will become more apparent to one of ordinary skill in the art to which the present disclosure pertains by referencing the detailed description of the present disclosure given below.

(7) According to an embodiment of the disclosure, an apparatus for manufacturing a display device, includes a stage; and an ejection head including an ejection outlet, which is open upward in a first direction. The ejection head is disposed under the stage and configured to move in a second direction, which intersects the first direction. “upward in the first direction” is opposite to a direction of gravity.

(8) In an embodiment, the ejection outlet may be disposed to face the stage and extend in a third direction, which intersects the first and second directions.

(9) In an embodiment, the ejection head may include a body part, which contains an adhesive material, and a stirring part, which is disposed in the body part and includes a roller.

(10) In an embodiment, the ejection head may further include an ejection driving part, which is connected to the body part and ejects the adhesive material to outside through the ejection outlet.

(11) In an embodiment, the ejection head may further include a heat unit disposed in the body part to surround the stirring part.

(12) In an embodiment, the apparatus may further include: a light irradiation unit disposed under the stage and which applies light toward a bottom surface of the stage, and the light irradiation unit may be configured to move in the second direction.

(13) In an embodiment, during an ejection process using the ejection head, the ejection head and the light irradiation unit may move in the second direction, and the light irradiation unit may be disposed in a direction opposite to the second direction from the ejection head.

(14) In an embodiment, during the ejection process using the ejection head, a distance between the ejection head and the light irradiation unit in the second direction may be constantly maintained.

(15) In an embodiment, the ejection head and the light irradiation unit may be integrated with each other.

(16) According to an embodiment of the disclosure, an apparatus for manufacturing a display device, includes: a stage including a mounting surface, to which a display panel is fixed, where the stage is rotatable, and an ejection head which ejects an adhesive material onto the mounting surface in a case where the stage is rotated such that the mounting surface faces downward in a first direction, which corresponds to that the mounting surface faces in a direction of gravity.

(17) In an embodiment, the stage may include a rotation unit, which changes an orientation direction of the mounting surface.

(18) In an embodiment, the ejection head may be disposed to face the mounting surface when the

mounting surface faces downward.

(19) In an embodiment, the ejection head may include an ejection outlet, which is open upward in the first direction and ejects the adhesive material, and “upward in the first direction” may be opposite to a direction of gravity.

(20) In an embodiment, the ejection outlet may extend in a second direction, which intersects the first direction, and the ejection head may eject the adhesive material while moving in a third direction, which intersects the first and second directions.

(21) In an embodiment, the apparatus may further include: a light irradiation unit which applies light to the display panel while moving in the third direction, when the ejection head is ejecting the adhesive material. The light irradiation unit may be disposed in a direction opposite to the third direction from the ejection head, and the light irradiation unit may temporarily cure the adhesive material, disposed on the mounting surface.

(22) In an embodiment, the display panel may define an opening therein, and the ejection head may be disposed under the display panel and eject the adhesive material while moving in a second direction, which intersects the first direction, but not eject the adhesive material while passing by the opening.

(23) In an embodiment, the display panel may include an application surface which faces the ejection head when the ejection head is ejecting the adhesive material, and the application surface may be more hydrophilic than a side surface of the display panel that defines the opening.

(24) According to an embodiment of the disclosure, a method of manufacturing a display device, includes: fixing a display panel on a stage; rotating the stage such that the display panel faces downward; applying an adhesive material on a bottom surface of the display panel; curing the adhesive material, applied on the display panel; and coupling a cover window onto the display panel.

(25) In an embodiment, the applying of the adhesive material may be performed using an ejection head, and the ejection head may eject the adhesive material onto the bottom surface of the display panel while moving.

(26) In an embodiment, the curing of the adhesive material may be performed using a light irradiation unit, and the light irradiation unit may apply light to the adhesive material applied on the display panel, while moving in the same direction as the ejection head.

(27) According to the aforementioned and other embodiments of the present disclosure, an optically clear adhesive (OCA) can be applied to a uniform thickness on surfaces of display panels having various planar shapes.

(28) Other features and embodiments may be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other embodiments and features of the present disclosure will become more apparent by describing in detail embodiments thereof with reference to the attached drawings, in which:

(2) FIG. 1 is a plan view of a display device according to an embodiment of the present disclosure;

(3) FIG. 2 is a cross-sectional view taken along line II-II' of FIG. 1;

(4) FIG. 3 is a perspective view of an apparatus for manufacturing a display device according to an embodiment of the present disclosure;

(5) FIG. 4 is a cross-sectional view of an apparatus of FIG. 3;

(6) FIG. 5 is a flowchart illustrating a method of manufacturing a display device according to an embodiment of the present disclosure;

(7) FIGS. 6 through 14 are schematic views illustrating the method of FIG. 5;

(8) FIG. 15 is a cross-sectional view of an apparatus for manufacturing a display device according to another embodiment of the present disclosure; and

(9) FIG. 16 is a cross-sectional view of an apparatus for manufacturing a display device according to still another embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

(11) It will also be understood that when a layer is referred to as being “on” another layer or substrate, it can be directly on the other layer or substrate, or intervening layers may also be present. The same reference numbers indicate the same components throughout the specification.

(12) It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a first element discussed below could be termed a second element without departing from the teachings of the present invention. Similarly, the second element could also be termed the first element.

(13) Each of the features of the various embodiments of the present disclosure may be combined or combined with each other, in part or in whole, and technically various interlocking and driving are possible. Each embodiment may be implemented independently of each other or may be implemented together in an association.

(14) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, “a,” “an,” “the,” and “at least one” do not denote a limitation of quantity, and are intended to include both the singular and plural, unless the context clearly indicates otherwise. For example, “an element” has the same meaning as “at least one element,” unless the context clearly indicates otherwise. “At least one” is not to be construed as limiting “a” or “an.” “Or” means “and/or.” As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. Embodiments of the present disclosure will hereinafter be described with reference to the accompanying drawings.

(15) FIG. 1 is a plan view of a display device according to an embodiment of the present disclosure. FIG. 2 is a cross-sectional view taken along line II-II' of FIG. 1.

(16) Referring to FIGS. 1 and 2, a display device DD may include a display area DA and a non-display area NDA. The display area DA may be an area in which an image is displayed, and the non-display area NDA may be an area in which no image is displayed. The display area DA may be positioned in the middle of the display device DD, and the non-display area NDA may be positioned on the periphery of the display area DA. For example, the non-display area NDA may surround the display area DA.

(17) The display device DD may include a display panel DP and a cover window CW, which is disposed on the display panel DP.

(18) The display panel DP may include elements and circuits for displaying an image, such as, for example, pixel circuits (e.g., switching elements) and self-light-emitting elements that emit light in accordance with the operation of the pixel circuits. In some embodiments, the self-light-emitting elements may include organic light-emitting diodes (OLEDs), quantum-dot light-emitting diodes

(“QLEDs”), inorganic material-based micro-light-emitting diodes (microLEDs), and/or inorganic material-based nano light-emitting diodes (nano LEDs) having a nano-scale width and/or length. The display panel DP may have a first surface facing the cover window CW and a second surface opposite to the first surface.

(19) The display panel DP may include an opening OP, which penetrates the display panel DP in a thickness direction, and a notch NC. The opening OP may be disposed in the display area DA, but the present disclosure is not limited thereto. Alternatively, the opening OP may be disposed in the non-display area NDA. The notch NC may be recessed downwardly from the top edge of the display panel DP in a plan view.

(20) The cover window CW may be disposed at the front of the display panel DP, may protect the display panel DP, and may transmit light emitted from the display panel DP therethrough. The cover window CW may include a rigid material such as glass, plastic, or quartz.

(21) The cover window CW may be disposed to overlap with the display panel DP and cover the front of the display panel DP. The cover window CW may have a similar planar shape to the display panel DP and have a larger size than the display panel DP. For example, both short sides of the cover window CW may protrude outwardly beyond the display panel DP. The cover window CW may be disposed to overlap with the opening OP and the notch NC of the display panel DP.

(22) The display device DD may further include an adhesive member AD, which is disposed between the display panel DP and the cover window CW. The adhesive member AD may couple the display panel DP and the cover window CW. The adhesive member AD may be an optically clear adhesive (OCA) or an optically clear resin (“OCR”).

(23) The adhesive member AD may not be disposed in a region where the opening OP of the display panel DP is disposed. The adhesive member AD may be disposed in the entire display panel DP except for the opening OP. The adhesive member may be disposed in a region where the display panel DP and the cover window CW are in contact with each other.

(24) The adhesive member AD may be formed on the first surface of the display panel DP by an apparatus **1** for manufacturing a display device according to an embodiment of the present disclosure, and the apparatus **1** will hereinafter be described with reference to FIGS. **3** and **4**.

(25) FIG. **3** is a perspective view of an apparatus for manufacturing a display device according to an embodiment of the present disclosure. FIG. **4** is a cross-sectional view of the apparatus of FIG. **3**.

(26) Referring to FIGS. **3** and **4**, the apparatus **1** may include a stage **100**, an ejection head **200**, and a light irradiation unit **300**.

(27) The stage **100** may include a mounting unit **110**, a rotation unit **120**, which rotates the mounting unit **110**, and a fixing unit **130**, which is disposed in the mounting unit **110**.

(28) The mounting unit **110** may provide space in which a target substrate is mounted. Here, the target substrate may be the display panel DP of FIGS. **1** and **2**. In the description that follows, the target substrate will be described as being the display panel DP. The mounting unit **110** may have a flat plate shape. The planar shape of the mounting unit **110** may be similar to the planar shape of the target substrate. For example, the mounting unit **110** may have a rectangular shape in a plan view, but the present disclosure is not limited thereto.

(29) The rotation unit **120** may rotate the mounting unit **110**. The rotation unit **120** may change the orientation directions (i.e., normal direction of the surface) of the surfaces of the mounting unit **110** by rotating the mounting unit **110**. For example, the rotation unit **120** may rotate the mounting unit **110** such that a first surface of the mounting unit **110** that is disposed on a third direction Z, of the mounting unit **110** may face an opposite direction of the third direction Z, of the mounting unit **110** and a second surface of the mounting unit **110** that is disposed on the opposite direction of the third direction Z of the mounting unit **110** may face the third direction Z of the mounting unit **110**. Here, the second surface of the mounting unit **110** is opposite to the first surface of the mounting unit **110**.

- (30) The rotation unit **120** may include a rotation axis that extends in one direction. FIG. 3 illustrates the rotation unit **120** as including a rotation axis extending in a second direction Y, but the present disclosure is not limited thereto. That is, the direction in which the rotation axis of the rotation unit **120** extends may be any direction parallel to the plane defined by a first direction X and the second direction Y.
- (31) Although not specifically illustrated, the rotation unit **120** may further include a rotation driver, which rotates the mounting unit **110** with respect to the rotation axis. For example, the rotation driver may include a motor.
- (32) However, the rotation unit **120** is merely exemplary. That is, the orientation directions of the first and second surfaces of the mounting unit **110** may be changed by an element other than the rotation unit **120**.
- (33) The fixing unit **130** may be disposed in the mounting unit **110**. The fixing unit **130** may fix the target substrate on the mounting unit **110**. The fixing unit **130** may include an electrostatic chuck. Specifically, power may be applied to the fixing unit **130**. The fixing unit **130** may generate an electrostatic force with the power applied thereto and may fix the target substrate to the mounting unit **110** with the electrostatic force. Alternatively, the fixing unit **130** may include a vacuum suction chuck and may fix the target substrate to the mounting unit **110** with the vacuum suction chuck. In this case, although not specifically illustrated, the fixing unit **130** may include a plurality of suction holes.
- (34) The ejection head **200** may eject an adhesive material AM. The ejection head **200** may apply the adhesive material AM onto a second surface, facing downward, of the target substrate, which is fixed on a second side, facing downward, of the stage **100**. The ejection head **200** may apply the adhesive material AM while moving in the first direction X. The ejection head **200** may include a body part **210**, an ejection driving part **220**, and a stirring part **230**. A moving speed V1 of the ejection head **200** may be constant, but the present disclosure is not limited thereto. Alternatively, the moving speed V1 of the ejection head **200** may change in the process of applying the adhesive material AM.
- (35) The body part **210** may include an inner space to accommodate the adhesive material AM. The body part **210** may extend in the second direction Y. The body part **210** may include a receiving part **212** and an ejection part **211**.
- (36) The receiving part **212** may accommodate the adhesive material AM. Specifically, the adhesive material AM may be accommodated in the inner space of the receiving part **212**. The ejection part **211** may be disposed on an upper part, in the third direction Z, of the receiving part **212**. The receiving part **212** and the ejection part **211** may be spatially connected to each other. The ejection part **211** may include an ejection outlet OL, which is open upward in the third direction Z. The ejection outlet OL may extend in the second direction Y. The ejection outlet OL may be disposed to face a second surface, facing downward, of the stage **100**. An inner width W of the ejection part **211** in the first direction X may gradually decrease in the third direction Z.
- (37) The ejection driving part **220** may be connected to the body part **210**. The ejection driving part **220** may deliver a driving force for ejecting the adhesive material AM from the inside of the body part **210** to the outside of the body part **210**. For example, the ejection driving part **220** may include a syringe or a pump.
- (38) The ejection driving part **220** may provide an ejection driving force that pushes the adhesive material AM out of the ejection part **211**. That is, the adhesive material AM may be ejected by the ejection driving force from the ejection driving part **220**.
- (39) The ejection driving part **220** may assist with the ejection of the adhesive material AM by the ejection part **211**. The adhesive material AM in the ejection part **211** may move toward the target substrate due to an attractive force toward the target substrate. As the adhesive material AM in the ejection part **211** moves toward the target substrate, the ejection driving part **220** may move the adhesive material AM in the receiving part **212** toward the ejection part **211**.

(40) The stirring part **230** may be disposed in the body part **210**. The stirring part **230** may include a roller, which extends in the second direction Y. The stirring part **230** may control the viscosity of the adhesive material AM with the rotation of the roller. For example, the roller of the stirring part **230** may rotate a predetermined number of times per unit time and may thus be able to maintain the viscosity of the adhesive material AM. To maintain the viscosity of the adhesive material AM to a certain value, the number of rotations per unit time of the roller may be raised as the viscosity of the adhesive material AM increases, and the number of rotations per unit time of the roller may be reduced as the viscosity of the adhesive material AM decreases.

(41) The ejection head **200** may be disposed on the second side, in the third direction Z, of the stage **100**. The ejection head **200** may eject the adhesive material AM while moving in the first direction X. The ejection head **200** may be disposed under the stage **100**, and as the ejection outlet OL of the ejection head **200** is open upward, the adhesive material AM, ejected from the ejection head **200**, may be applied onto the second surface (i.e., surface facing downward) of the target substrate, fixed on a bottom surface of the stage **100**.

(42) The application of the adhesive material AM onto the target substrate with the use of the ejection head **200** may be performed upward. That is, the adhesive material AM may be ejected and applied onto the target substrate in the opposite direction of the direction of gravity. The direction of gravity may refer to the “second side in the third direction Z”, and the opposite direction of the direction of gravity may refer to the “first side in the third direction Z”. Thus, in a case where the target substrate defines an opening (see “OP” of FIG. 1) therein, the adhesive material AM may not be applied on a side surface of the target substrate that defines the opening OP. Also, in this case, the adhesive material AM may not be applied to portion of the first surface of the stage **100**, exposed by the opening OP. Thus, the stage **100** can be effectively prevented from being contaminated with the adhesive material AM.

(43) The ejection outlet OL may extend in the second direction Y. Thus, in a case where the ejection head **200** applies the adhesive material AM while moving in the first direction X, the adhesive material AM may be applied to a predetermined area of the target substrate, defined by the first and second directions X and Y. For example, if the ejection outlet OL has a first length in the second direction Y and the ejection head **200** ejects the adhesive material AM while moving in the first direction X by as much as a second length, the adhesive material AM may be applied to a rectangular area on the target substrate that has the second length in the first direction X and the first length in the second direction Y.

(44) The thickness to which the adhesive material AM is applied onto the target substrate may be controlled with the moving speed V1 of the ejection head **200**. For example, the faster the moving speed V1 of the ejection head **200** is, the thinner the adhesive material AM is applied onto the target substrate, and the slower the moving speed V1 of the ejection head **200** is, the thicker the adhesive material AM is applied onto the target substrate. That is, in a case where the viscosity of the adhesive material AM is uniform and the moving speed V1 of the ejection head **200** is constant, the thickness to which the adhesive material AM is applied onto the target substrate may also be uniform throughout the target substrate.

(45) The light irradiation unit **300** may be disposed on a second side, in the first direction X, of the ejection head **200**. The light irradiation unit **300** may temporarily cure the adhesive material AM by applying light to the adhesive material AM, applied onto the target substrate. The light irradiation unit **300** may be disposed in the opposite direction of the moving direction of the ejection head **200**.

(46) The light irradiation unit **300** may apply the light L upward. The light L may be ultraviolet light. The light L may temporarily cure the adhesive material AM, applied on the target substrate.

(47) The light irradiation unit **300** may apply the light L while moving in the first direction X. The moving direction of the light irradiation unit **300** may be the same as the moving direction of the ejection head **200**. Specifically, the light irradiation unit **300** may apply the light while moving in

the first direction X, from the second side, in the first direction X, of the ejection head **200**, when the ejection head **200** applies the adhesive material AM while moving in the first direction X. That is, to temporarily cure the adhesive material AM, applied by the ejection head **200**, the light irradiation unit **300** may apply the light L while following the ejection head **200**, from the second side, in the first direction X (or the moving direction), of the ejection head **200**.

(48) A moving speed V2 of the light irradiation unit **300** may be substantially the same as the moving speed V1 of the ejection head **200**. In this case, a distance d between the ejection head **200** and the light irradiation unit **300** in the first direction X may be constantly maintained during the application of the adhesive material AM. However, the present disclosure is not limited to this. Alternatively, the moving speed V2 of the light irradiation unit **300** may be lower or higher than the moving speed V1 of the ejection head **200**. The amount of light L applied by the light irradiation unit **300** may be raised as the moving speed V2 of the light irradiation unit **300** decreases, and the amount of light L applied by the light irradiation unit **300** may be reduced as the moving speed V2 of the light irradiation unit **300** increases.

(49) The light irradiation unit **300** may extend in the second direction Y. The light L applied by the light irradiation unit **300** may be applied in the form of linear light extending in the second direction Y, but the present disclosure is not limited thereto. Alternatively, the light L applied by the light irradiation unit **300** may extend in the second direction Y, but may be in the form of surface light having a predetermined width in the first direction X.

(50) During the application of the adhesive material AM and the application of the light L, the distance d between the light irradiation unit **300** and the ejection head **200** in the first direction X may be constantly maintained. Specifically, the interval between the application of the adhesive material AM, ejected from the ejection head **200**, onto the target substrate and the application of the light L by the light irradiation unit **300** may become uniform.

(51) The apparatus **1** can apply the adhesive material AM to a uniform thickness on the target substrate, using the ejection head **200**, which includes the ejection outlet OL extending in the second direction Y and ejects the adhesive material AM while moving in the first direction X.

(52) Also, the apparatus **1** includes the ejection head **200**, which includes the ejection outlet OL that is open upward, and can thus apply the adhesive material AM onto the second surface, in the third direction Z, of the target substrate, fixed on the second side, in the third direction Z, of the stage **100**. That is, the apparatus **1** can apply the adhesive material AM in the opposite direction of the direction of gravity. Thus, the adhesive material AM may not move toward the stage **100**. That is, the stage **100** can be effectively prevented from being contaminated with the adhesive material AM. Also, in a case where the target substrate includes the opening OP, the side surface of the target substrate that defines the opening OP can be effectively prevented from being contaminated with the adhesive material AM.

(53) FIG. 5 is a flowchart illustrating a method of manufacturing a display device according to an embodiment of the present disclosure. FIGS. 6 through 14 are schematic views illustrating the method of FIG. 5.

(54) A target substrate on a stage **100** will hereinafter be described as being, for example, a display panel DP, and for convenience, the display panel DP is illustrated in FIGS. 6 through 14 as defining an opening OP therein, but not including a notch NC.

(55) Referring to FIGS. 5 and 6, the method of manufacturing a display device may include fixing the display panel DP on the stage **100** (S11), rotating the stage **100** such that the display panel DP faces downward (S21), forming an adhesive member AD by applying an adhesive material AM on a first surface of the display panel DP with the use of an ejection head **200** and temporarily curing the adhesive material AM (S31), and attaching a cover window CW on the display panel DP (S41).

(56) The display panel DP may be fixed on the stage **100** (S11). The display panel DP may define the opening OP therein. As already mentioned above with reference to FIGS. 3 and 4, the stage **100** may include a mounting unit **110**, a rotation unit **120**, and a fixing unit **130**. The display panel DP

may be disposed on the mounting unit **110** of the stage **100**. Specifically, the display panel DP may be disposed on a first surface (i.e., a mounting surface) of the mounting unit **110**.

(57) Once the display panel DP is disposed on the first surface of the mounting unit **110**, the fixing unit **130** is driven to fix the display panel DP on the mounting unit **110**. Here, the expression “the fixing unit **130** is driven” means that power is supplied to the fixing unit **130** in a case where the fixing unit **130** is, for example, an electrostatic chuck, or negative pressure is provided to the fixing unit **130** in a case where the fixing unit **130** is, for example, a vacuum suction chuck.

(58) Thereafter, referring to FIGS. 5, 6, and 7, the stage **100** may be rotated such that the display panel DP may face downward (S21).

(59) The stage **100** may be rotated by the rotation unit **120**. The stage **100** may be rotated such that the first surface of the display panel DP faces downward. The rotation unit **120** may have a rotation axis extending in the second direction Y and may rotate the stage **100** about the rotation axis. As a result of the rotation of the stage **100**, a facing direction of the first surface of the mounting unit **110** and a facing direction of a second surface of the mounting unit **110** may be interchanged. Here, the second surface of the mounting unit **110** is opposite to the first surface of the mounting unit **110**. That is, the stage **100** may be rotated 180 degrees. Even though the stage **100** is rotated, the display panel DP may be fixed by the fixing unit **130**.

(60) In response to the stage **100** being rotated, the display panel DP, fixed by the mounting unit **110**, may face downward (i.e., face a direction opposite to the direction of the gravity). If the display panel DP is not fixed by the fixing unit **130**, the display panel DP may fall from the stage **100** due to gravity. However, as the display panel DP is fixed to the mounting unit **110** by the fixing unit **130**, the display panel DP may not be detached from the stage **100**.

(61) Thereafter, referring to FIGS. 5 and 8 through 12, the adhesive member AD is formed (S31) by applying the adhesive material AM on the first surface of the display panel DP with the use of the ejection head **200** and temporarily curing the adhesive material AM.

(62) FIG. 8 illustrates how to apply and temporarily cure the adhesive material AM on an application surface DPb of the display panel DP with the use of the ejection head **200** and a light irradiation unit **300**. FIGS. 9 through 11 illustrate how to form the adhesive member AD through the application of the adhesive material AM around the opening OP. FIG. 12 illustrates the adhesive member AD as being formed on the display panel DP.

(63) The ejection head **200** may apply the adhesive material AM on the application surface DPb of the display panel DP while moving in the first direction X. The application surface DPb of the display panel DP may be disposed to face the second side in the third direction Z (i.e., downward). The adhesive material AM may be ejected from an ejection outlet OL of the ejection head **200** and may be applied on the application surface DPb of the display panel DP. The ejection head **200** may apply the adhesive material AM on the application surface DPb of the display panel DP while not being in contact with the display panel DP. A moving speed V1 of the ejection head **200** may be constant. Thus, the adhesive material AM may be applied with a uniform thickness on the application surface DPb of the display panel DP, and as a result, the thickness of the adhesive member AD may be uniform throughout the display panel DP.

(64) The application surface DPb of the display panel DP may be hydrophilically treated with plasma. As a result, the application surface DPb of the display panel DP may be more hydrophilic than the side surfaces of the display panel DP. The adhesive material AM, which is applied onto the display panel DP, may be hydrophilic. An attractive force may act between the adhesive material AM and the display panel DP. Thus, the adhesive material AM can be stably applied onto the application surface DPb of the display panel DP.

(65) The ejection outlet OL of the ejection head **200** may be placed near the application surface DPb of the display panel DP, and the adhesive material AM may be ejected from the ejection head **200** to move toward the application surface DPb of the display panel DP. Then, the adhesive material AM may be attached on the application surface DPb of the display panel DP.

(66) Referring to FIG. 10, when the ejection head **200** is passing by a region where the opening OP is defined, the adhesive material AM may become relatively distant from the application surface DPb of the display panel DP and thus may not be ejected. That is, in a case where the ejection head **200** is placed to overlap with the opening OP in the third direction Z (i.e., in a plan view), the adhesive material AM may not be ejected from the ejection head **200**. Also, the adhesive material AM can be effectively prevented from moving toward a side surface of the display panel DP that defines the opening OP. Thus, the adhesive material AM may not be applied on the side surface of the display panel DP that defines the opening OP.

(67) Also, the adhesive material AM may not be applied on portion of the surface of the stage **100** that is exposed by the opening OP of the display panel DP. As the adhesive material AM is applied to the display panel DP turned upside down, the adhesive material AM can be effectively prevented from entering the opening OP due to gravity. As the application surface DPb of the display panel DP is hydrophilically treated but the side surface of the display panel DP that defines the opening OP is not hydrophilically treated, the adhesive material AM may not enter the opening OP. Also, when the ejection outlet OL overlaps the opening OP in a plan view, as the ejection outlet OL and the application surface DPb of the display panel DP become relatively distant from each other, the adhesive material AM may not be ejected. Thus, even if the application of the adhesive material AM is repeatedly performed, the side surface of the display panel DP that defines the opening OP and the portion of the surface of the stage **100** that is exposed by the opening OP of the display panel DP can be effectively prevented from being contaminated with the adhesive material AM.

(68) As illustrated in FIG. 11, in a case where the ejection head **200** is placed back near the application surface DPb of the display panel DP (i.e., the ejection head **200** is placed back to overlap the application surface DPb) after passing by the region where the opening OP is disposed, the adhesive material AM may be ejected from the ejection outlet OL and may then be applied onto the application surface DPb of the display panel DP.

(69) The ejection head **200** may apply the adhesive material AM while moving in the first direction X, and at the same time, the light irradiation unit **300** may apply light L toward the first side in the third direction Z (i.e., upward) while moving from a second side to a first side (e.g., from left to right) in the first direction X. A moving speed V2 of the light irradiation unit **300** may be substantially the same as the moving speed V1 of the ejection head **200**, but the present disclosure is not limited thereto. Alternatively, the moving speed V2 of the light irradiation unit **300** may be lower or higher than the moving speed V1 of the ejection head **200**. In a case where the moving speed V1 of the ejection head **200** and the moving speed V2 of the light irradiation unit **300** are the same, the distance between the ejection head **200** and the light irradiation unit **300** may be constant.

(70) The light irradiation unit **300** may temporarily cure the adhesive material AM by applying the light L. In this manner, the adhesive member AD may be formed. Here, the expression “the adhesive material AM is temporarily cured” means that the adhesive material AM is not completely cured, but only half cured. The adhesive member AD, formed on the display panel DP, may have adhesiveness and may thus be coupled to the cover window CW later. The distance between the light irradiation unit **300** and the ejection head **200** in the first direction X may be constantly maintained.

(71) As the adhesive material AM is applied on the application surface DPb of the display panel DP and is then temporarily cured by applying the light L, the adhesive member AD may be formed on the display panel DP.

(72) Thereafter, referring to FIGS. 5, 13, and 14, the cover window CW may be attached on the display panel DP (**S41**).

(73) The attachment of the cover window CW may be performed after rotating the stage **100** again, but the present disclosure is not limited thereto. Alternatively, the attachment of the cover window CW may be performed with the display panel DP fixed on a second side, in the third direction Z, of the stage **100**. It will hereinafter be described how to attach the cover window CW on the display

panel DP after rotating the stage **100** again such that the display panel DP is placed back to be oriented so as to face upward.

(74) Once the adhesive member AD is formed on the application surface DPb of the display panel DP, the stage **100** may be rotated again by the rotation unit **120**.

(75) The cover window CW may be coupled onto a first surface, in the third direction Z, of the display panel DP. The cover window CW may be coupled to the display panel DP by the adhesive member AD, which is disposed on the first surface, in the third direction Z, of the display panel DP. That is, the adhesive member AD may be disposed between the cover window CW and the display panel DP, but not in the region where the opening OP is disposed.

(76) Apparatuses for manufacturing a display device according to other embodiments of the present disclosure will hereinafter be described, focusing mainly on the differences with the apparatus **1**.

(77) FIG. **15** is a cross-sectional view of an apparatus for manufacturing a display device according to another embodiment of the present disclosure.

(78) Referring to FIG. **15**, an apparatus **1_1** for manufacturing a display device DD differs from the apparatus **1** in that an ejection head **200** and a light irradiation unit **300** are connected and thus integrated with each other.

(79) The apparatus **1_1** may further include a coupling unit **400**, which is disposed between the ejection head **200** and the light irradiation unit **300**. The coupling unit **400** may integrate the ejection head **200** and the light irradiation unit **300** with each other. A first end, in the first direction X, of the coupling unit **400** may be coupled to the ejection head **200**, and a second end, in the first direction X, of the coupling unit **400** may be coupled to the light irradiation unit **300**. Thus, the ejection head **200** and the light irradiation unit **300** may move together. Also, the distance between the ejection head **200** and the light irradiation unit **300** in the first direction X may be constantly maintained. A distance d between the ejection head **200** and the light irradiation unit **300** may be substantially the same as the length of the coupling unit **400**.

(80) The apparatus **1_1** includes the ejection head **200**, which includes an ejection outlet OL extending in the second direction Y, and may apply an adhesive material AM to a uniform thickness onto a target substrate with the use of the ejection head **200**, which ejects the adhesive material AM while moving in the first direction X.

(81) The apparatus **1_1** includes the ejection head **200**, which includes an ejection outlet OL that is open upward, and may apply the adhesive material AM onto a second surface, in the third direction Z, of the target substrate, fixed on a second side, in the third direction Z, of a stage **100**. That is, the apparatus **1_1** may apply the adhesive material AM in the opposite direction of the direction of gravity. Thus, the adhesive material AM may not move toward the stage **100**. In other words, the stage **100** can be effectively prevented from being contaminated with the adhesive material AM. Also, in a case where the target substrate includes an opening OP, a side surface of the target substrate that defines the opening OP can be effectively prevented from being contaminated with the adhesive material AM.

(82) Also, as the apparatus **1_1** further includes the coupling unit **400**, which integrates the ejection head **200** and the light irradiation unit **300** with each other, the distance between the ejection head **200** and the light irradiation unit **300** in the first direction X can be constantly maintained. Thus, an adhesive member AD can be formed with a uniform degree of curing.

(83) FIG. **16** is a cross-sectional view of an apparatus for manufacturing a display device according to still another embodiment of the present disclosure.

(84) Referring to FIG. **16**, an apparatus **1_2** for manufacturing a display device DD differs from the apparatus **1** in that an ejection head **200_2** further includes a heater unit **240**, which is disposed in a body part **210** of the ejection head **200_2**.

(85) The ejection head **200_2** may further include the heater unit **240**, which is disposed in the body part **210** of the ejection head **200_2**. The heater unit **240** may be disposed along the inner sidewall of the body part **210**. The heater unit **240** may be disposed to surround a stirring part **230**.

The heater unit **240** may generate heat and may thus control the temperature and the viscosity of a material contained in the body part **210**. For example, the heater unit **240** may control the temperature and the viscosity of an adhesive material AM, contained in the body part **210**, by supplying heat to the adhesive material AM. As the temperature of the adhesive material AM increases, the viscosity of the adhesive material AM decreases, and as the temperature of the adhesive material AM decreases, the viscosity of the adhesive material AM decreases. Thus, the viscosity of the adhesive material AM can be effectively controlled not only using a roller of the stirring part **230**, but also using heat provided by the heater unit **240**.

(86) The apparatus **1_2** includes the ejection head **200_2**, which includes an ejection outlet OL extending in the second direction Y and ejects the adhesive material AM while moving in the first direction X, and may apply the adhesive material AM to a uniform thickness onto a target substrate.

(87) The apparatus **1_2** may apply the adhesive material AM to a uniform thickness onto the target substrate, using the ejection head **200_2**, which includes an ejection outlet OL that is open upward, and may apply the adhesive material AM onto a second surface, in the third direction Z, of the target substrate, fixed on a second side, in the third direction Z, of a stage **100**. That is, the apparatus **1_2** may apply the adhesive material AM in the opposite direction of the direction of gravity. Thus, the adhesive material AM may not move toward the stage **100**. In other words, the stage **100** can be effectively prevented from being contaminated with the adhesive material AM. Also, in a case where the target substrate includes an opening OP, a side surface of the target substrate that defines the opening OP can be effectively prevented from being contaminated with the adhesive material AM.

(88) Also, as the apparatus **1_2** can control the viscosity of the adhesive material AM, contained in the body part **210**, by controlling the temperature of the adhesive material AM with the use of the heater unit **240**, which is disposed in the body part **210** of the ejection head **200_2**.

(89) In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications can be made to the preferred embodiments without substantially departing from the principles of the present invention. Therefore, the disclosed preferred embodiments of the invention are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. An apparatus for manufacturing a display device, comprising: a stage; and an ejection head including an ejection outlet, which is open in a first direction opposite to a direction of gravity, wherein the ejection head is configured to move in a second direction, which intersects the first direction, and disposed under the stage in the direction of the gravity, wherein the ejection head includes a body part, which contains an adhesive material, and a stirring part, which is disposed in the body part and includes a roller, wherein the roller is configured to raise the number of rotations per unit time thereof when viscosity of the adhesive material increases and configured to reduce the number of rotations per unit time thereof when the viscosity of the adhesive material decreases so as to maintain the viscosity of the adhesive material to a certain value, wherein the adhesive material is hydrophilic, wherein the ejection head further includes a heat unit disposed along an inner sidewall of the body part to surround the stirring part, the heat unit is spaced apart from the stirring part with the adhesive material therebetween, an inner width of an upper part of the heat unit in the second direction gradually decreases in the first direction, the heat unit includes an opening corresponding to the ejection outlet, and the adhesive material is discharged through the ejection outlet and the opening, wherein the heat unit is in direct contact with the adhesive material.
2. The apparatus of claim 1, wherein the ejection outlet is disposed to face the stage and extends in a third direction, which intersects the first and second directions.
3. The apparatus of claim 1, wherein the ejection head further includes an ejection driving part,

which is connected to the body part and ejects the adhesive material to outside through the ejection outlet.

4. The apparatus of claim 1, further comprising: a light irradiation unit disposed on the opposite direction of the first direction, from the stage, the light irradiation unit applying light toward a second surface, in the opposite direction of the first direction, of the stage, wherein the light irradiation unit is configured to move in the second direction.

5. The apparatus of claim 4, wherein during an ejection process using the ejection head, the ejection head and the light irradiation unit move in the second direction, and the light irradiation unit is disposed in a direction opposite to the second direction from the ejection head.

6. The apparatus of claim 5, wherein during the ejection process using the ejection head, a distance between the ejection head and the light irradiation unit in the second direction is constantly maintained.

7. The apparatus of claim 4, wherein the ejection head and the light irradiation unit are integrated with each other.

8. An apparatus for manufacturing a display device, comprising: a stage including a mounting surface, to which a display panel is fixed, wherein the stage is rotatable; and an ejection head which ejects an adhesive material onto the mounting surface in a case where the stage is rotated such that the mounting surface faces in an opposite direction of a first direction, wherein the first direction is an opposite direction of gravity, wherein the ejection head includes a body part, which contains an adhesive material, and a stirring part, which is disposed in the body part and includes a roller, wherein the roller is configured to raise the number of rotations per unit time thereof when viscosity of the adhesive material increases and is configured to reduce the number of rotations per unit time thereof when the viscosity of the adhesive material decreases so as to maintain the viscosity of the adhesive material to a certain value, wherein the adhesive material is hydrophilic, wherein the ejection head further includes a heat unit disposed along an inner sidewall of the body part to surround the stirring part, the heat unit is spaced apart from the stirring part with the adhesive material therebetween, an inner width of an upper part of the heat unit in the second direction gradually decreases in the first direction, the heat unit includes a first opening corresponding to the ejection outlet, and the adhesive material is discharged through the ejection outlet and the first opening, wherein the heat unit is in direct contact with the adhesive material.

9. The apparatus of claim 8, wherein the stage includes a rotation unit, which changes an orientation direction of the mounting surface.

10. The apparatus of claim 8, wherein the ejection head is disposed to face the mounting surface when the mounting surface faces in a direction of gravity.

11. The apparatus of claim 8, wherein the ejection head includes an ejection outlet, which is open in the first direction and ejects the adhesive material.

12. The apparatus of claim 11, wherein the ejection outlet extends in a second direction, which intersects the first direction, and the ejection head ejects the adhesive material while moving in a third direction, which intersects the first and second directions.

13. The apparatus of claim 12, further comprising: a light irradiation unit which applies light to the display panel while moving in the third direction, when the ejection head is ejecting the adhesive material, wherein the light irradiation unit is disposed in a direction opposite to the third direction from the ejection head, and the light irradiation unit temporarily cures the adhesive material disposed on the mounting surface.

14. The apparatus of claim 8, wherein the display panel defines a second opening therein, and the ejection head is disposed under the display panel and ejects the adhesive material while moving in a third direction, which intersects the first direction, but does not eject the adhesive material while passing by the second opening.

15. The apparatus of claim 14, wherein the display panel includes an application surface which

faces the ejection head when the ejection head is ejecting the adhesive material, and the application surface is more hydrophilic than a side surface of the display panel that defines the second opening.
